Andhra Pradesh State Road Transport Corporation(APSRTC)
Visakhapatnam City Bus Transit Analytics -Case Study

Inference: APSRTC Vizag has recovered fully from COVID with 8L daily passengers in FY24, but the EPKM–CPKM gap (₹37.6 vs ₹41.2) means every km operated loses ₹3.6.

Inference: BS6 buses - 374 (59% of fleet), BS3 buses -35 (need immediate replacement), 2018 procurement of 118 buses will cross the 15-year threshold by 2033. EV replacement is the optimum solution for these buses

Inference: Maddilapalem needs a dedicated investigation — it has the second-highest route count (22) and good fleet utilisation (91.8%) but the OTP (82.9%) is less compared to all other depots.

Inference: Routes 38IT,1T has maximum passengers and revenue. whereas routes routes 25M and 5D has good revenue similar in lines to routes 38IT,1T. Increasing number of buses in these routes increases revenue to government as well services to the public

PREPARED BY

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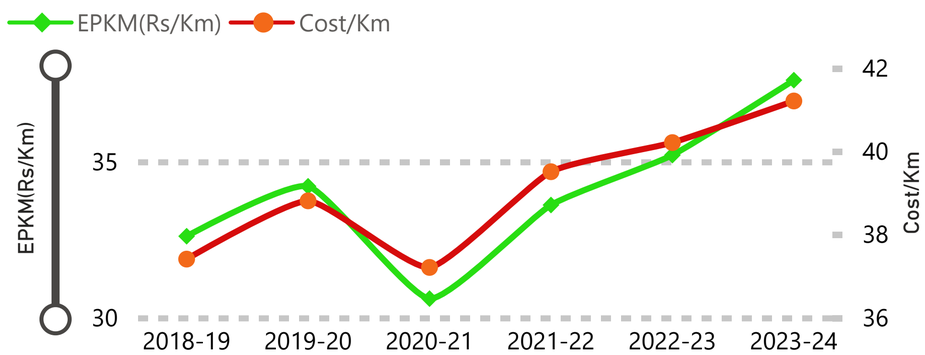
APSRTC Dashboard-Executive

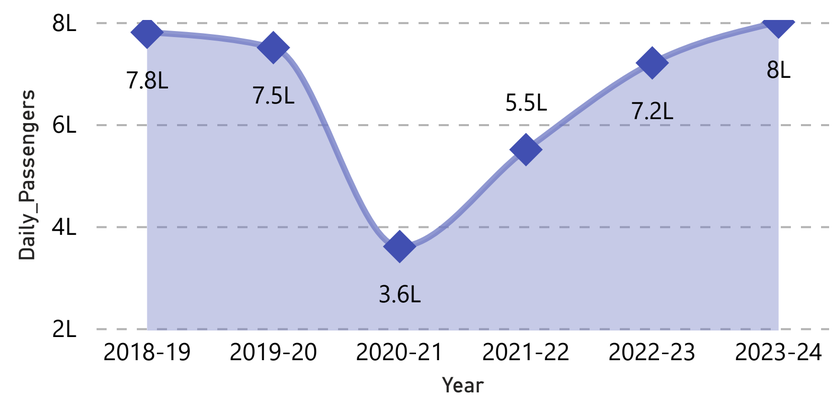
2018-19	2019-20	2020-21	2021-22	2022-23	2023-24

670	84	8.0L	2.20L	37.60	78.0L	92.69	78.2
Total City Buses	Active Routes	Daily_Passengers	KM Operated/Day	EPKM(Rs/Km)	Daily Revenue	Fleet Utilization%	Arrival On-Time Performance%

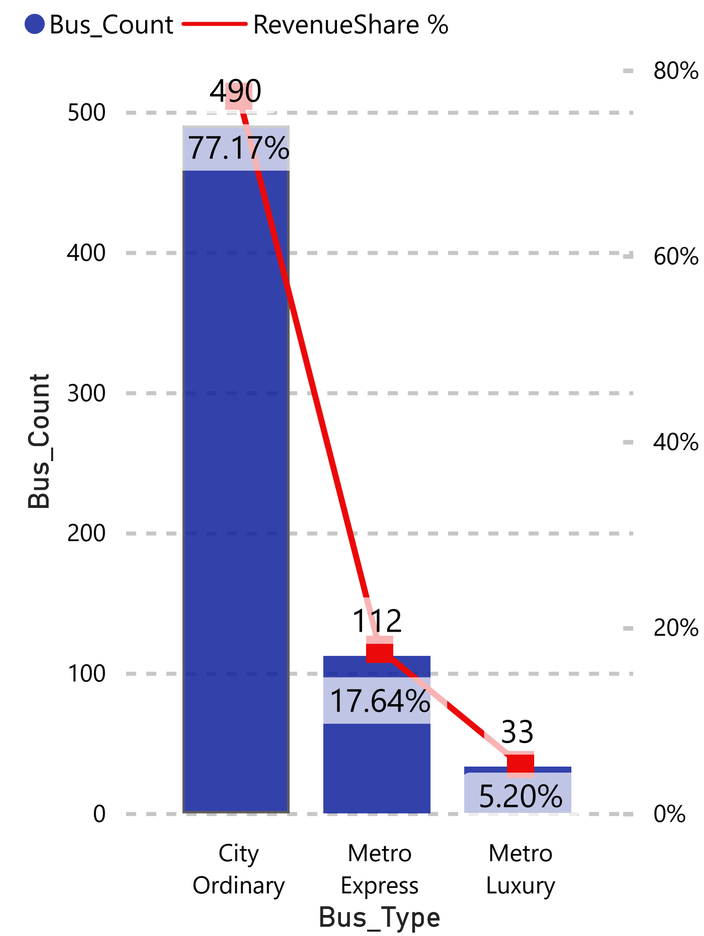
Revenue vs Cost/Km (Rs/Km)



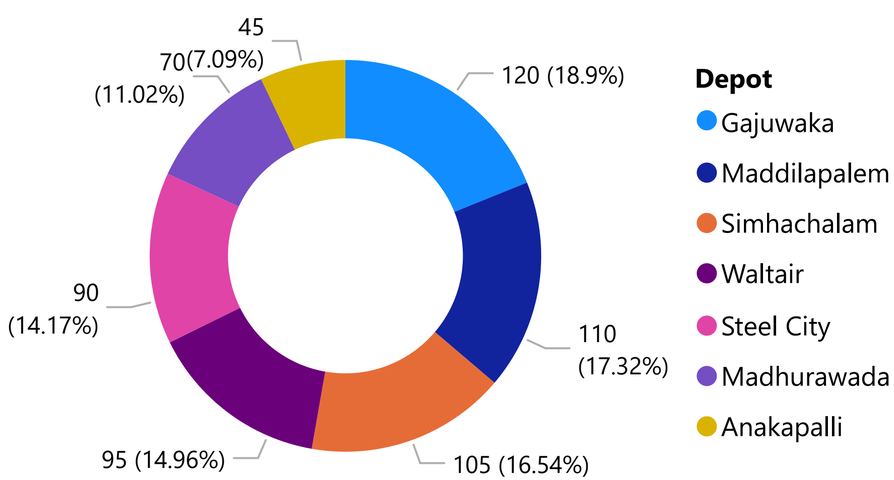
Daily_Passengers Trend (Lakh)



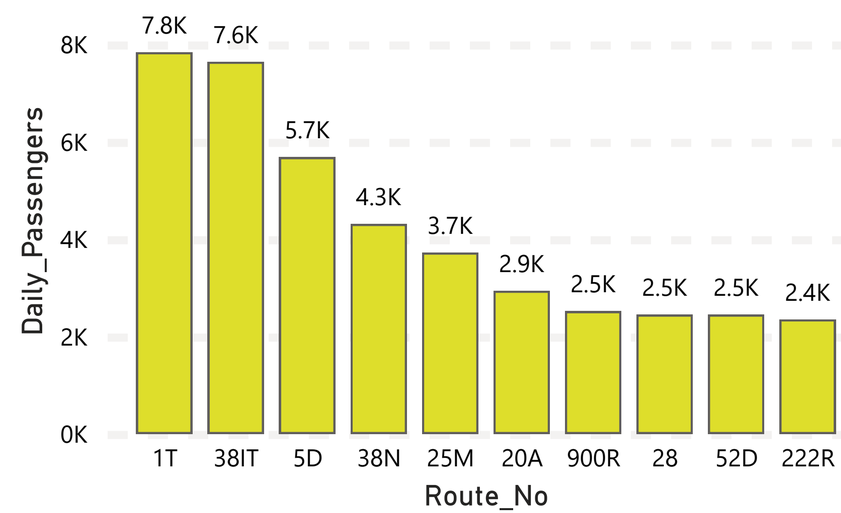
Bus_Type-Fleet Vs RevenueShare %



Fleet by Depot (635 Buses,7Depots,Reserve buses-35)



Top 10 Routes - Daily Passengers



APSRTC Dashboard-Fleet

2018-19	2019-20	2020-21	2021-22	2022-23	2023-24

670 Total City Buses

49 Maint_Buses

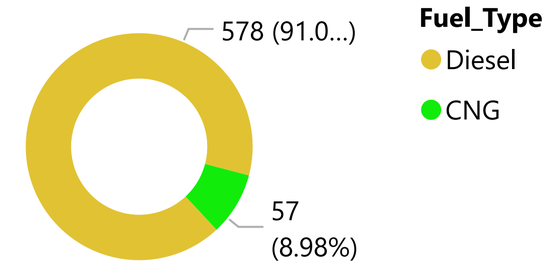
621 Effective Fleet Operatng

316 Km/Bus/Day (Avg)

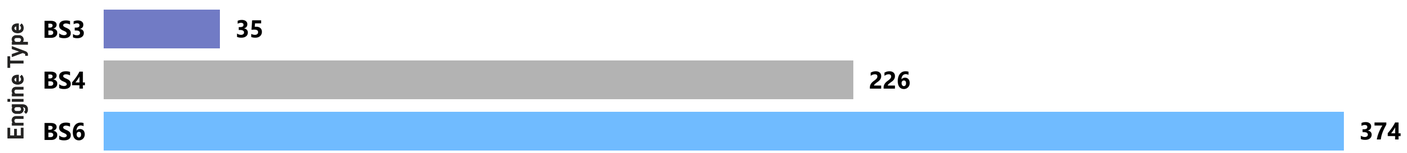
12580 Revenue/bus/day(rs)

64 Fleet_age>10 yrs

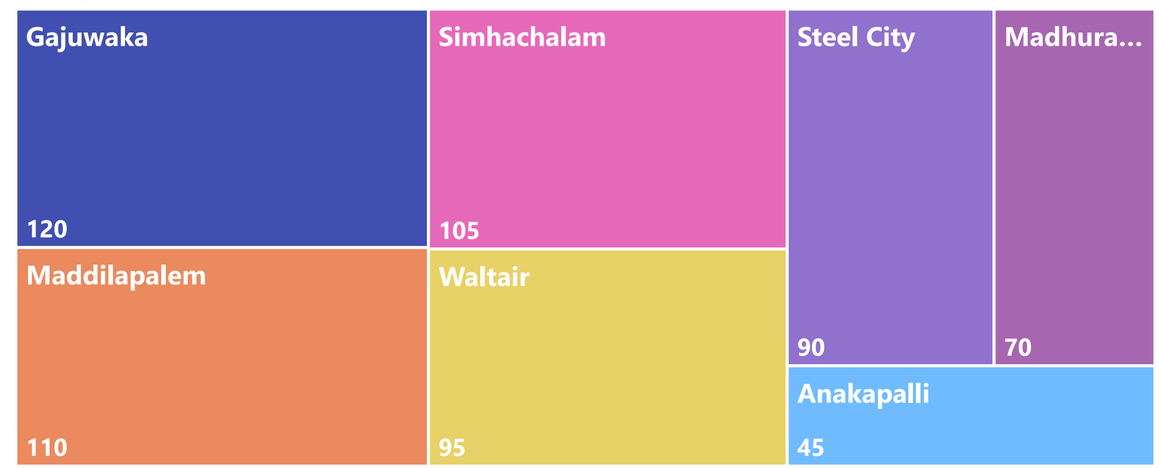
Buses_By Fuel Type



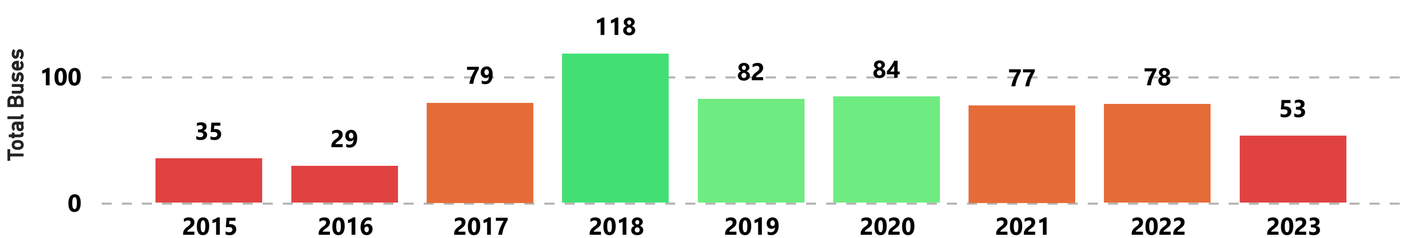
Buses_by Engine Type



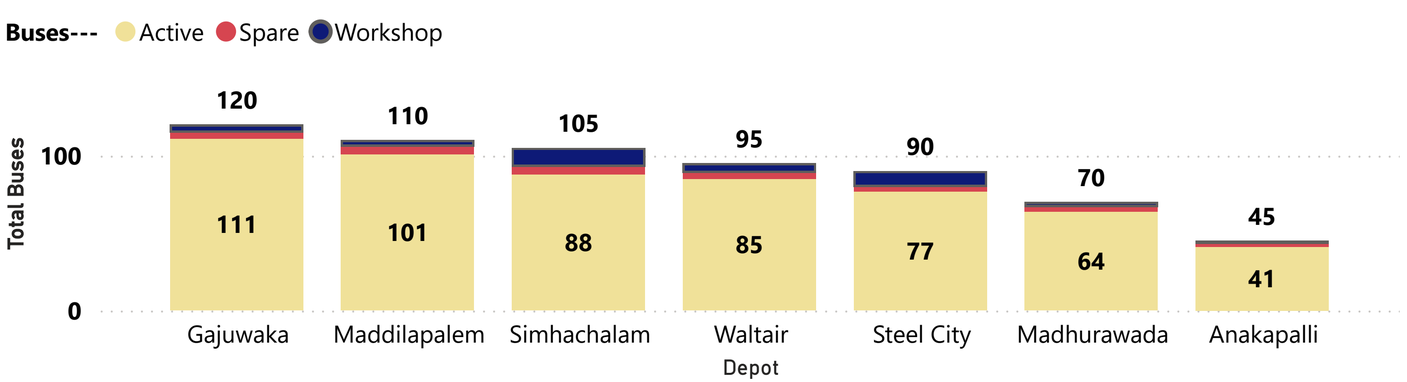
GPS_Fitted Buses/Depot



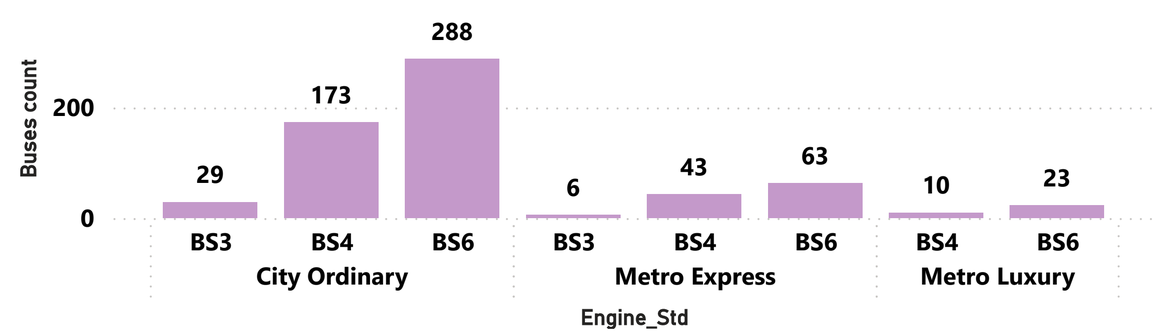
Buses_Manufactured/Year



Depot's Buses status (Active/Spare/Workshop)



Buses Types with Engine Type

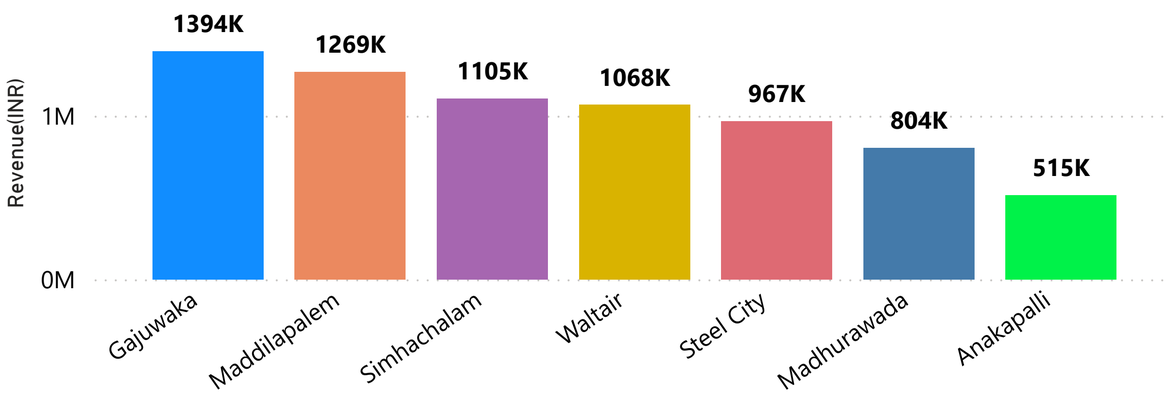


APSRTC Dashboard - Depot Score card

2018-19	2019-20	2020-21	2021-22	2022-23	2023-24

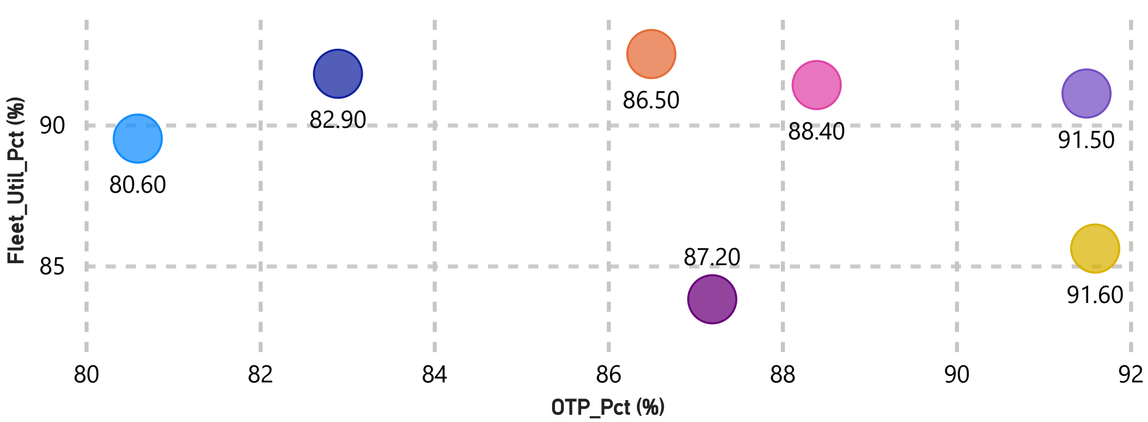
7	450	94.20	91.30	41.20	1.1	15.34	122.22
Bus Depots	Bus Stops / Shelters	Service Reliability %	Fare Box Recovery Ratio (%)	Cost/Km	Fleet CAGR %	EPKM Growth %	Pax_Recovery %(Post-Covid)

Revenue_Depots_Wise

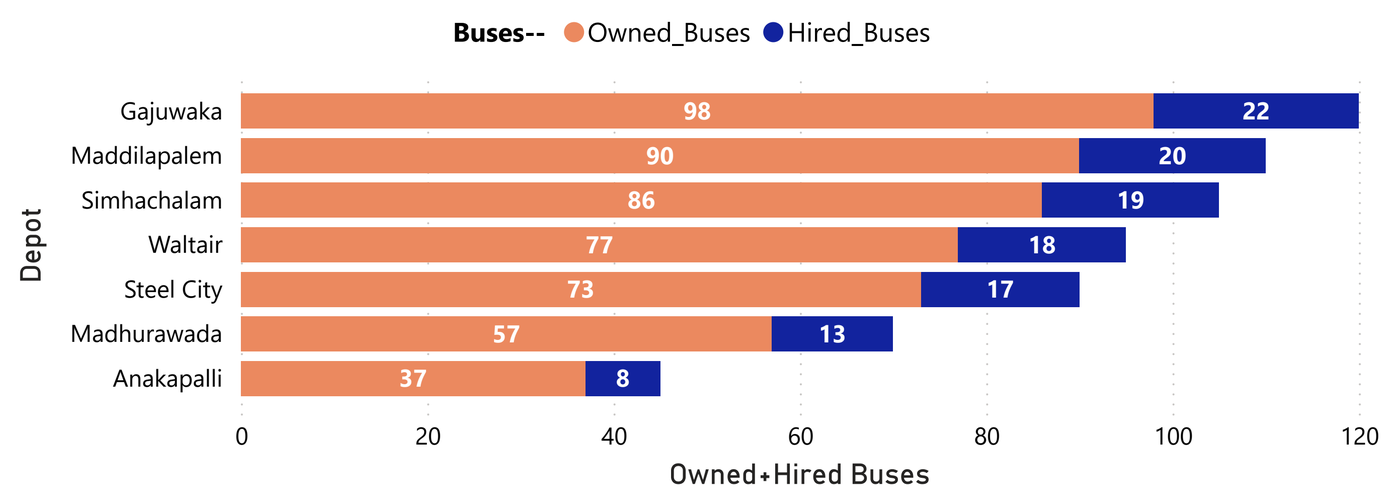


Depot	Total_Buses	Routes_Operated	Daily_Pax_Boarding Count	OTP_Pct	Fleet_Util_Pct	RevenueShare %	Daily_Revenue_INR
Gajuwaka	120	14	143K	86.50	92.50	18.90%	1394K
Maddilapalem	110	22	130K	82.90	91.80	17.32%	1269K
Simhachalam	105	19	113K	87.20	83.80	16.54%	1105K
Waltair	95	11	110K	80.60	89.50	14.96%	1068K
Steel City	90	7	99K	91.60	85.60	14.17%	967K
Madhurawada	70	6	82K	88.40	91.40	11.02%	804K
Anakapalli	45	5	53K	91.50	91.10	7.09%	515K

Fleet_Util_Pct vs OTP_Pct



Depots vs Buses (Own + Hired)



APSRTC Dashboard-Routes, Revenue & Operations

2018-19	2019-20	2020-21	2021-22	2022-23	2023-24

98 GPS / AIS-140 Fitted Buses(%)

82.00 CO2 Emissions per Pax-KM (gms)

91.20 ePOS Digital Ticketing %

2,384 Total Trips

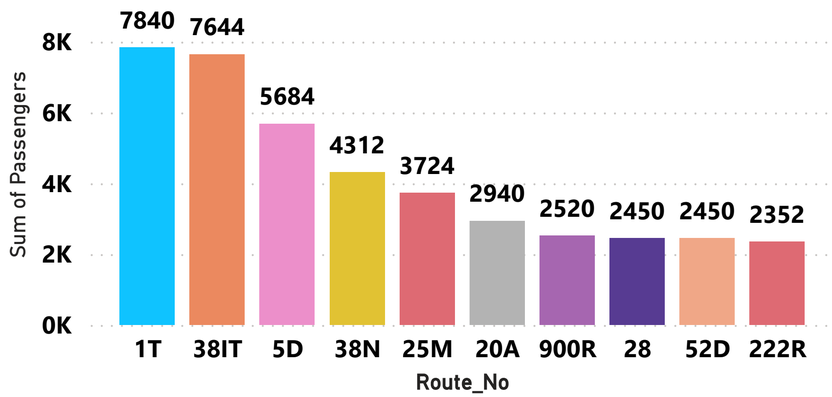
94.09 Departure_Punctuality %

2.9% Breakdown rate

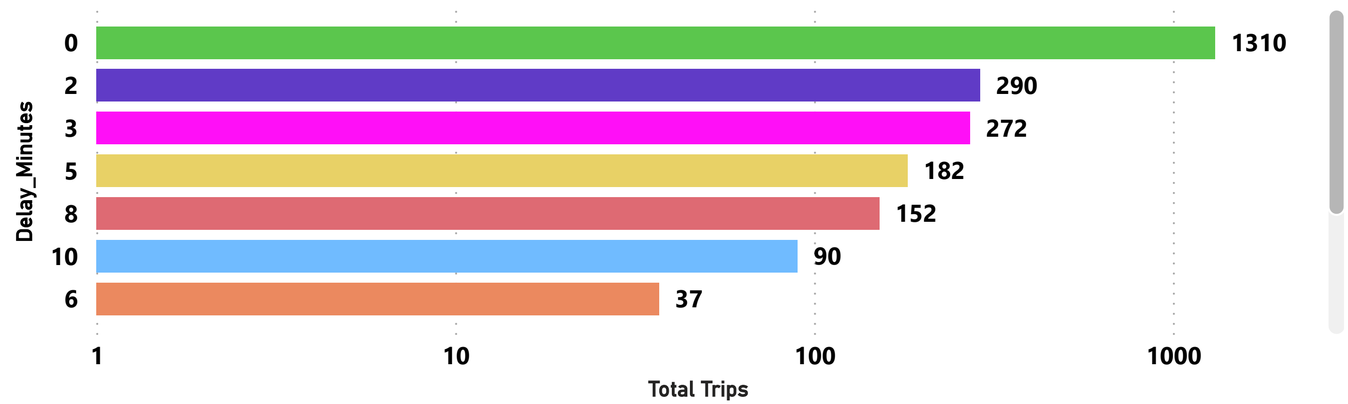
2.27 Avg Trip Delay(min)

2243 Trips on Time

Top 10 Routes_Passengers Count

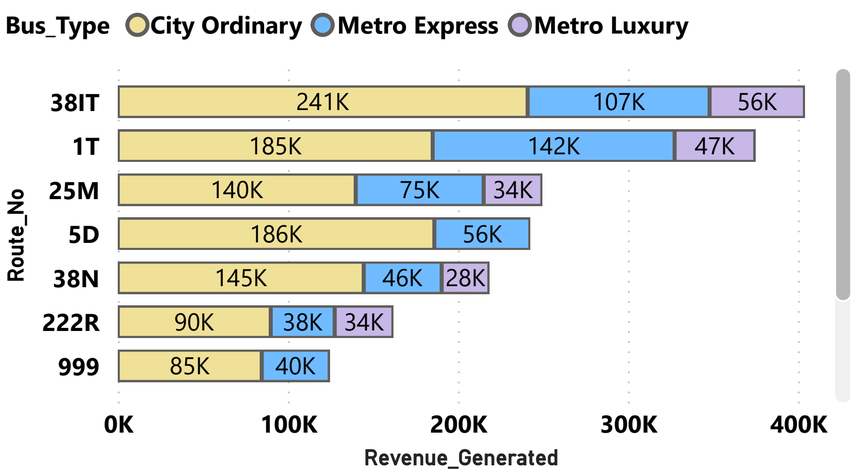


Trips vs Delay_Min

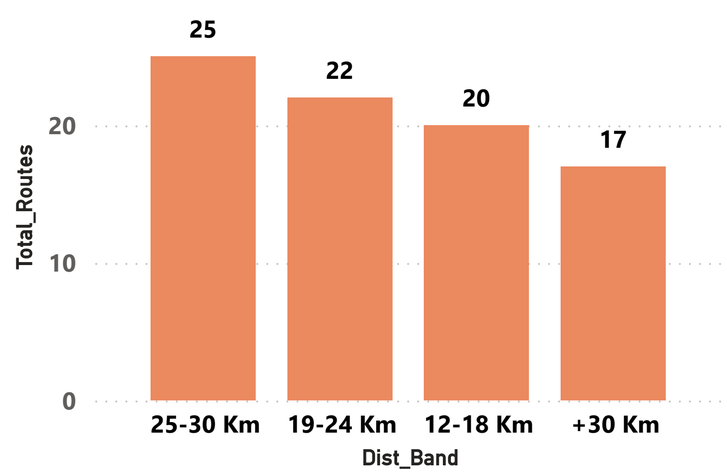
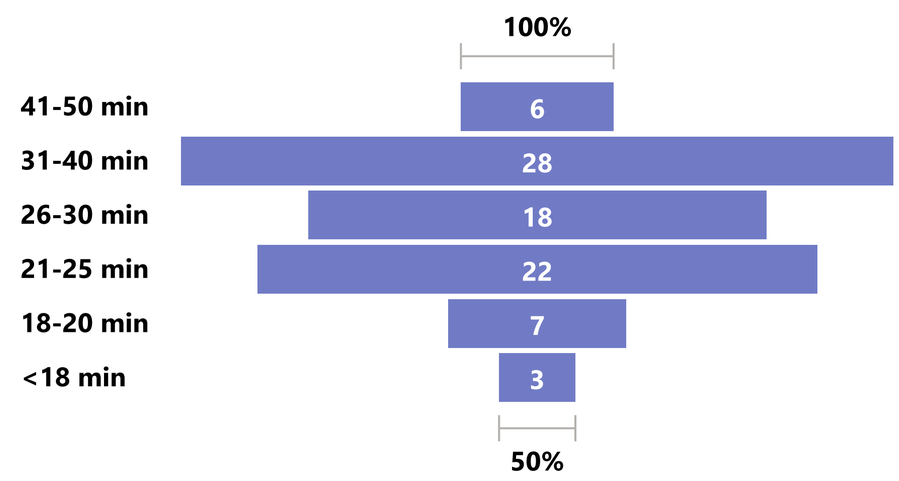


25 Avg_Route_Length(km)

Top 10 Routes_Revenue Generation



Routes Coverage vs Distance



Routes Coverage vs Time